Advanced Lab Report:
Characterizing the Red Line Noise Curve

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Abstract

The telescope dome on top of Cudahy Science Hall is currently without a telescope due to the vibrational noise from the Red Line trains which pass closely by, roughly every five minutes. In this work, we design and implement an experiment to characterize the excess noise signal generated by these trains, working towards the future implantation of a new telescope. We construct simple seismometers to detect the vibrational signal, which is then analyzed using power spectral density estimates. These estimates show that there is a clearly detectable increase in seismic noise underneath the Red Line, although we fail to detect any significant signal at Cudahy Science Hall. We consider implications of our results, how our experimental design choices influenced these outcomes, and how future experiments may improve upon our methods.

I. INTRODUCTION AND MOTIVATION

The physics department of Loyola University Chicago (LUC), located in Cudahy Science Hall (CSH), has no working astronomical telescope installed in the dome which sits atop the building. From conversation with faculty in the department [3], we have learned that this is in part due to the vibrations from the nearby Chicago Transit Authority (CTA) Red Line train. Specifically, this vibrational noise would constantly throw off the delicate calibration required for imaging of distant bodies unless an additional, expensive suspension system was installed to mitigate it. Our goal in this experiment is to characterize this train-generated seismic signal, towards the development and installation of a working telescope.

This vibrational issue is not exclusive to LUC. For example, the LIGO observatories must account for noise from nearby train lines that affect the gravitational wave detectors' data [4]. It is possible to similarly identify and account for vibrational noise produced by the CTA train line [1] that reaches CSH. Similar vibration studies have been performed before, leading us to believe that light-rail induced vibrations will still be a dominant signal even at distance from the train line [2]. The CTA measures above 50 dB (relative to 1μinch/s) within a frequency band spanning $\approx 10-40$ Hz at distances of up to 300 feet [1]. Our experiment plans to nearly double this distance, making the analysis of any true train signal detected in CSH an important data point for constructions similarly close to the CTA elevated trains.

We are also motivated by the potential our experiment has to enrich the education of future LUC physics students. The most likely form of this is through the continuation of our experiment, even in spirit. Either through the use of our code and detectors, or through simply following our line of questioning, we sincerely hope that this experiment is further pursued in the future. Another, admittedly less likely, outcome of this experiment is that we do lay the ground work for some kind of telescope to be installed at CSH. The science and educational benefit to future students in this case is trivial.

We structure this paper as follows: In section II, we cover the design and construction of the seismometer network. In section III, we discuss the data collection, cleaning, and analysis methods used to produce our results, which are presented in section IV. Finally, in section V, we draw physical conclusions about our results and potential systematic error introduced by the design of our seismometer network.

Component	Amount	Estimated Price
Arduino Uno R4 Wifi	3	$82.50
Arduino GPS Shield	3	$89.85
GPS Antenna	3	$64.50
Antenna Adapter Cable	3	$11.85
Micro SD Card (256 GB)	3	$239.97
Portable Power Bank	3	$41.97
Plastic Jar	3	$18.99
30 Gauge Magnet Wire	3	$38.94
Neo. Magnet Set	1	$8.99
Spring Set	1	$7.89
Threaded Rod Set	1	$7.59
Total (Retail)	N/A	$613.04
Total (actual)	N/A	$530.54

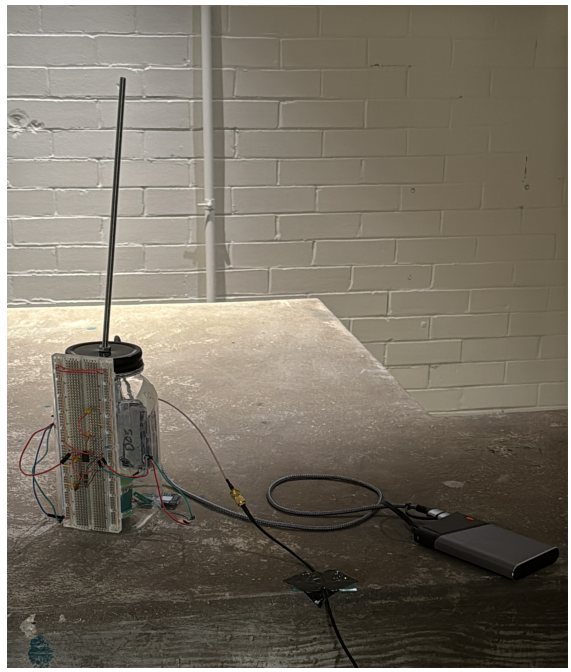

FIG. 1: *(left)* Price summary for required components of the experiment. Note that prices vary based on manufacturer and retailer. *(right)* A photo of one seismometer on the LUC Physics Department's seismic monolith.

II. DESIGN

A. Materials and Equipment

To detect vibrations and convert them into workable data, we needed a number of materials. The Arduino components and other items we purchased to build our detector are listed in figure 1. In addition to these components, we also require the parts to make the amplifier circuit described in figure 2, which include the circuit components, wiring, breadboard, and soldering iron.

After constructing our detectors and performing our experimental runs, we see a number of flaws in our design present in our choice of materials. A first and trivial issue with our current materials list is the fact that we have chosen plastic jars. Their lighter weight, combined with the flexibility of the material, likely reduced the power of the signal traveling from the bottom of the jar to our spring and magnets. We will discuss further how choices for our detector materials may

have impacted our ability to detect seismic signals in section V.

B. The Detectors

We describe the design and construction of our three, independent seismometers based around the Arduino Uno R4 Wifi microcontroller, called Uno, Dos, and Tres. To convert the motion of the ground into an initial voltage, we first constructed a set of three geophones loosely following the design outlined in Mirko Pavleski's 2024 *Maker.io* blog post [7]. In short, we suspend a stack of neodymium magnets from a spring above a coil of copper magnet wire. The plastic jars act as a housing for this setup, allowing for some insulation from outside influence and keeping each detector in a single (relatively) contained package. We also encase the coil of copper wire in aluminum foil, to both reduce outside electromagnetic interference and to help reduce unwanted resonance of the magnet through eddy currents.

To the outside of the geophone housing we attached the amplifier circuit and Arduino stack, again to keep these detectors compact. The Arduino stack consists of the Uno R4 microcontroller and Adafruit Ultimate GPS Logger Shield mentioned in figure 1. The Arduino was responsible for providing 5 volt power, ground, and an analog sensor pin, while running the data logger program and pre-processing our data. The GPS logger pulled accurate time and location for each detector in order to keep our data aligned, and also handled the writing of our data to micro SD cards in the form of CSV files.

The amplifier circuit is based around the circuit described by K&J Magnetics in reference [6]. The main components of this circuit (pictured in figure 2) are the coil, feedback loop, 5 volt source, bias voltage, and output. The coil is coupled to the circuit through a capacitor (C6) to prevent DC noise, and raised to the bias voltage of 2.5 volts in order to avoid clipping near the power rails of our LT1001 op amp. The resistor and capacitor in the feedback loop (R1 and C5 in the diagram) control the gain of the amplifier, which then sends the boosted signal out to the analog sensor pin on the Arduino stack.

The removable top of the jar allowed us to run the detectors without a magnet, which was helpful in characterizing the expected output. We observed a noise curve in the output, even when running with no magnet to generate the signal, which is representative of the inherent noise in our amplifier circuit. When performing these runs with no magnet, our amplifiers produced a noise curve (figure 3) which is similar to the inherent noise described in the LT1001 datasheet [8].

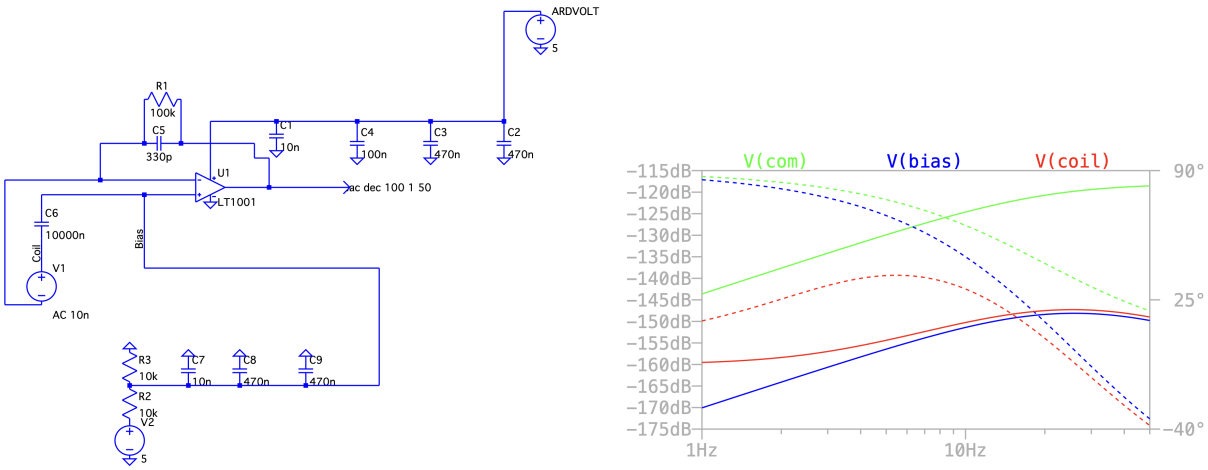

FIG. 2: Circuit diagram of amplifier circuit (left). Theoretical amplification of the circuit from LTspice (right). The blue curve is our bias voltage, the red curve is the raised signal from our coil, and the green curve is the output from the amplifier.

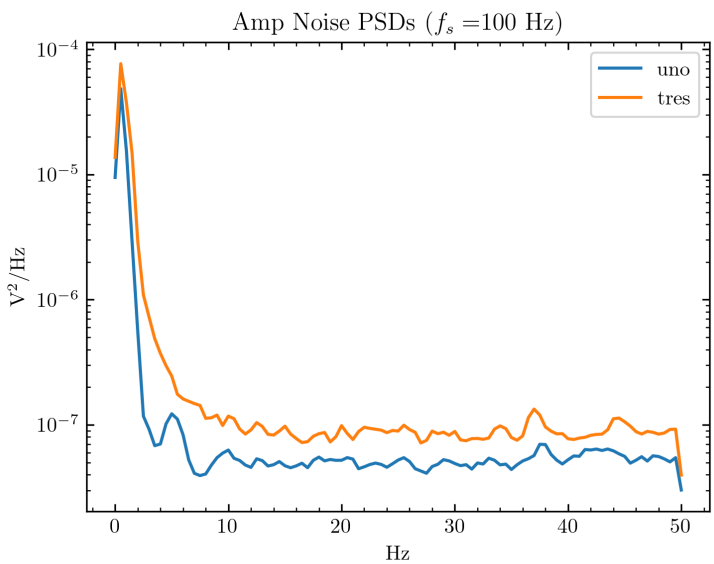

FIG. 3: A PSD of the data collected by detectors without their magnets, indicating detector noise.

Namely, we see significant noise below 10 Hz which prevents meaningful probes into this regime. While the LT1001 exhibits "low enough" noise for most applications, we will see that the noise floors in figure 3 inhibit our ability to detect seismic signals at distance later.

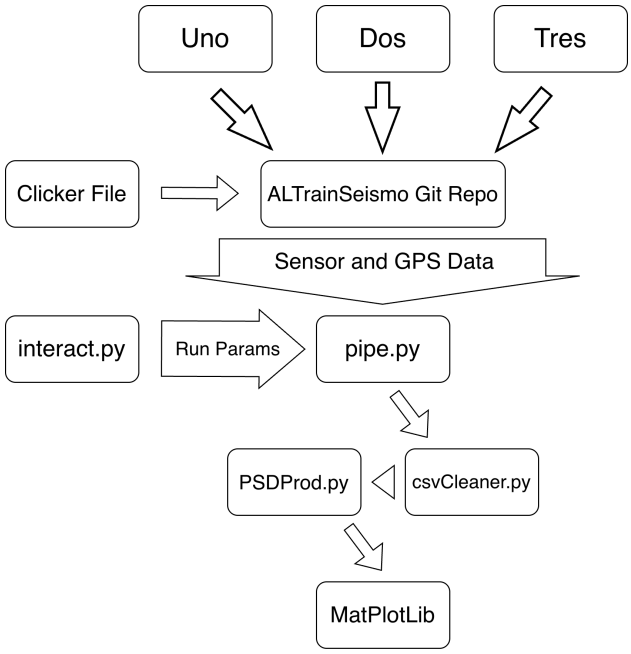

FIG. 4: Flowchart for our analysis pipeline.

III. METHODS

With the magnets placed back into the detectors they were distributed to three separate locations: Uno was placed in a LUC utilities room underneath the Red Line, Dos on the seismic monolith at CSH, and Tres in the room directly below the dome of CSH. After starting the detectors, around 1:30 PM on Wednesday April 15, we manually recorded when trains were either stopping at or passing by the train station. Once the observing run had finished, around 3:30 PM, we exported the data off of the SD cards and onto our GitHub repository for analysis [5]. We also uploaded a CSV file with event timestamps to the GitHub repository, so we can manually keep track of when trains were passing over Uno.

Our analysis pipeline (summarized in figure 4), begins by taking manually inputted parameters and using these to clean the data from the detectors. Specifically, we align start and stop times for each detector, and then apply windowing to isolate event times from background times (see figure 5). Due to the distance between our location for Uno and the other detectors, the event timestamps we manually record may not be valid for Dos and Tres. In this case, we apply a statistical windowing to their data, assigning an event if the signal voltage is in the extreme 0.1%. We can then compare this statistical windowing to the manually identified events to see how accurate our automated windows are.

We then used the Welch's method function included in the Python package SciPy [9] to compute an estimate of the power spectral density (PSD) for each window. This function takes a time series signal and breaks it up into overlapping segments, to which it applies a window function. Then, it computes the magnitude squared of the discrete Fourier transform of each segment, normalizing by the width of the segment. These results are then averaged to estimate the power density at each frequency, producing a PSD estimate. A PSD estimate, in plain terms, describes the strength of some signal in terms of its frequency components. In signal analysis, these facilitate the recognition of resonant or dominant modes in a signal, which is especially useful for analysis of noise signals. For a more detailed explanation, see Welch's original paper in reference [10].

The final signal analysis will need to consider the strength of the train event signal over the background noise. For this, we must convert from units of V^2/Hz to decibels (dB). As a refresher, for some characteristic power P_0 and signal power P, the signal strength in decibels is defined in equation 1. Using PSDs from detector runs with no magnets as our characteristic power allows us to determine the strength of any signal above this noise floor. Using the runs with no magnets as our floor allows us to determine the sensitivity to only the signal in a frequency dependent manner.

$$dB = 10 \times \log\left(\frac{P}{P_0}\right) \tag{1}$$

IV. RESULTS

Our concurrent run of the seismometer network produced approximately two hours of vibrational data, which we present in figures 5 and 6. As expected, we observed a much larger signal from the detector placed under the Red Line (Uno) than the one placed directly under the dome at CSH (Tres). Regretfully, the detector placed on CSH's seismic monolith (Dos) malfunctioned during this concurrent run and we could not regain access to the locations we needed to place Uno and Tres.

In figure 5 we see just how much stronger the train signal is detected by Uno than by Tres, as Uno's voltage varies by ≈ 3 volts and Tres' voltage varies by only ≈ 0.1 volts. We also notice that there is a sever lack of event windows for Tres in late times, where Uno has some of its most significant detections. In figure 6, we see differing significant spikes in the signal power between Uno and Tres for the noise, event, and background decibel PSD plots. These show up in the plotted train signal strength as dips in the signal power, because decibels here are defined in relation to the

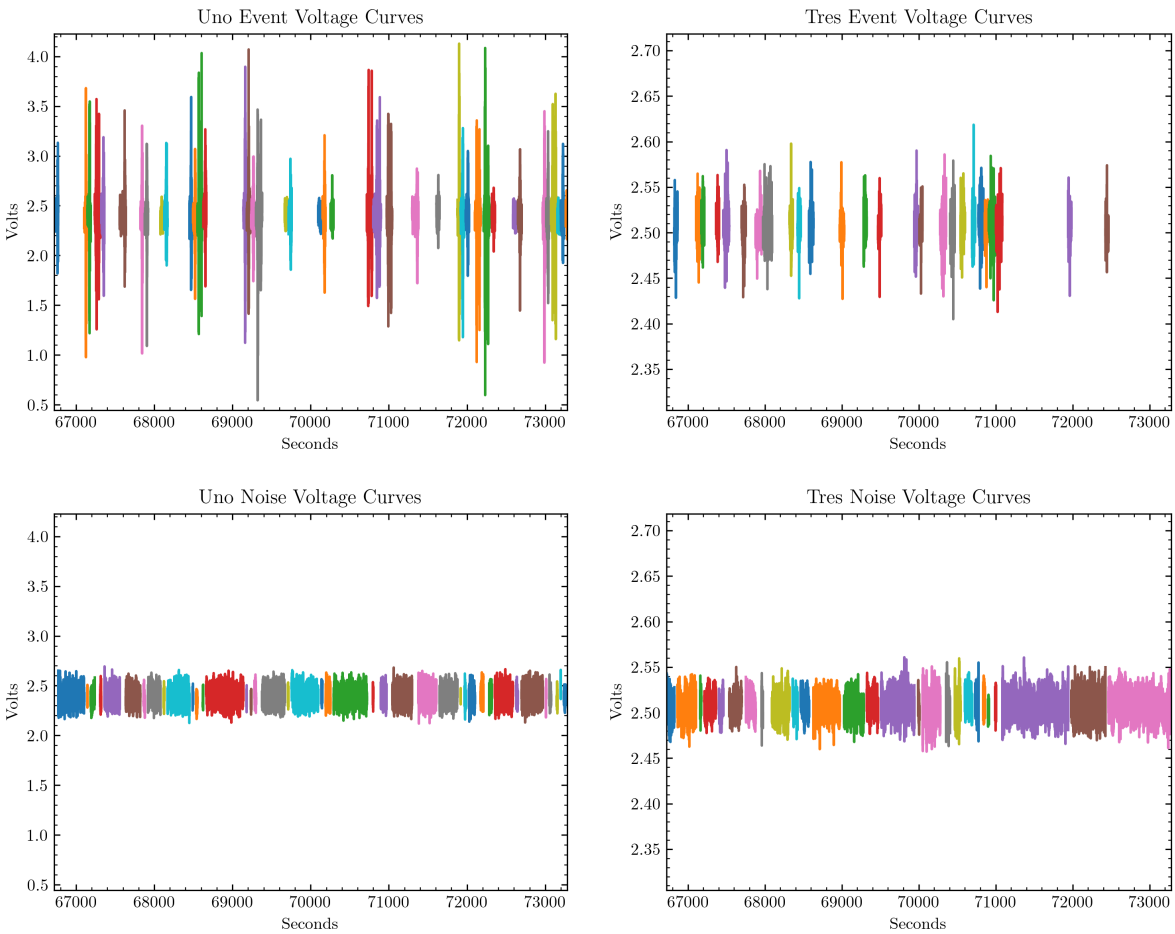

FIG. 5: Windowed time series for detectors Uno and Tres.

background noise.

We also perform analysis of the signal-to-noise ratio (SNR) across our frequency band of interest (10-50 Hz), which is plotted against the event window half-width in figure 7. The raw data and scripts used to produce these plots can be found in our project's GitHub repository [5].

V. CONCLUSIONS

Reviewing our data and PSD plots, we can see that Uno is clearly detecting the additional train-induced seismic noise. A peak, frequency band wide SNR of 13.13 dB ($\approx$ 20.6 linear SNR) combined with the manually recorded train event timestamps significantly support this conclusion, showing it is in fact possible to characterize the Red Line vibrational signal. This is an exciting result and is encouraging for the future of seismology at LUC. However, we believe that Tres is

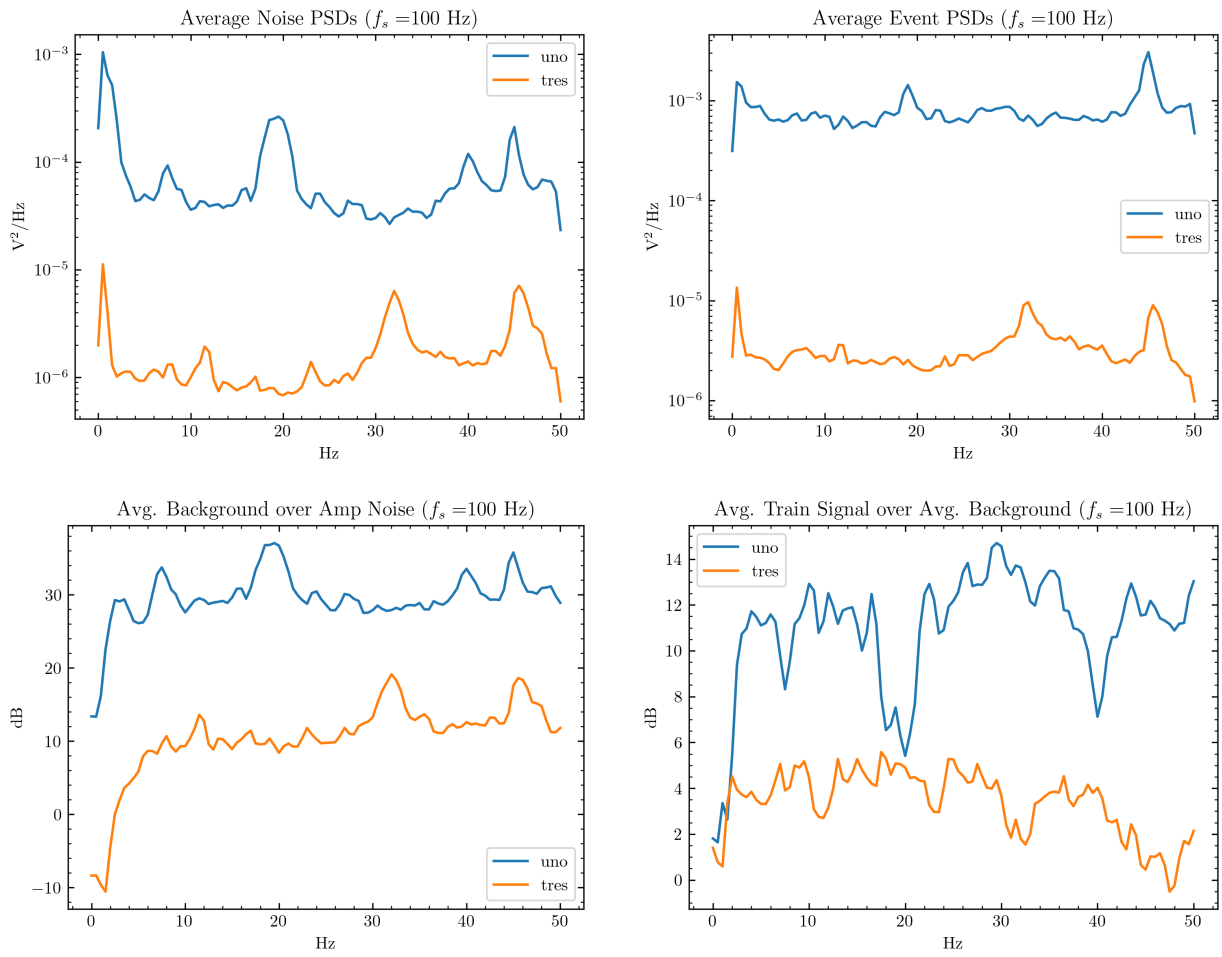

FIG. 6: Average PSD plots of background windows (*top left*), event windows (*top right*), background noise above amp noise (*bottom left*), and train signal above background noise (*bottom right*).

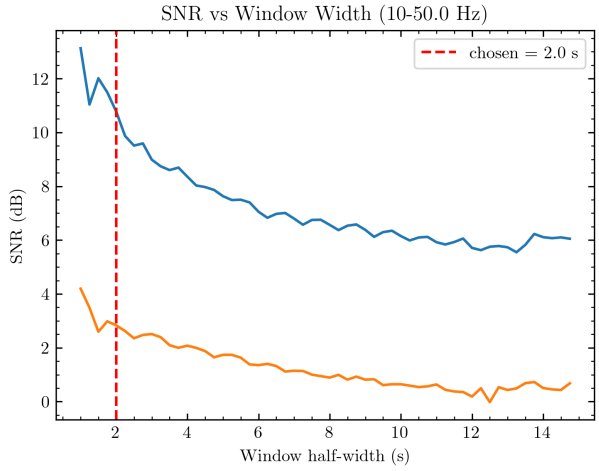

FIG. 7: The signal-to-noise ratio graphs, which show a peak around 13 dB in Uno and a peak around 4 dB in Tres.

not hearing any distinguishable train signal over the generic noise floor. Tres' SNR peaks at 4.20 dB (≈ 2.6 linear SNR), which is not significant enough to suggest any true detection. We also see that the event signal in Tres' PSD plots is not consistent with the frequency band of Uno's events (10-50 Hz).

This is further supported by the poor alignment between Uno's and Tres' event windowing. We expect windows between Uno and Dos/Tres to shift by about a minute, due to the Loyola Red Line stop being some distance away from the Red Line's point of closest approach to CSH. That Tres hears essentially no events in late times (where Uno hears many) suggests that our statistical windowing is not capturing any true events, except by chance. This makes sense when we consider how strong the signal *should* be at CSH.

If we assume that Uno and Tres are identical detectors, and that the dissipation of the Red Line's seismic signal follows an inverse square law, we can perform a back of the napkin calculation to predict the post-amplification signal strength at CSH. Knowing that CSH is approximately 170 meters from the Red Line, we can say that $\approx 0.0035\%$ of the signal present underneath the Red Line should make it to CSH. Using $10^{-3}\ V^2/\text{Hz}$ as a characteristic signal strength for Uno's detected events, we see that a train signal at CSH should have a strength around $3 \times 10^{-8}\ V^2/\text{Hz}$. This is below the amplifier noise floor of Tres and thus completely undetectable.[1]

As Dos malfunctioned, we cannot draw any conclusions about how detecting from the seismic monolith would impact our findings. However, as all three detectors are built to be the same, we expect the results would look similar to Tres'.

While we do not see a significant signal in Tres, there are several methods to improve the probability that a seismometer placed inside CSH will detect significant vibrational signal due to the nearby train. In a future iteration of this experiment, we would prioritize constructing spring-magnet systems with resonant frequencies squarely in the center of our frequency band of study. This would increase the vibrational power transferred through the magnet to the coil and thus increase our baseline signal strength. We would also order more sensitive components for the amplifier to try and lower the noise floor inherent to our detectors. Finally, we would work to increase the coupling of our detectors to the ground either by increasing detector weight or by some method of attachment.

In all, although we fail to detect the Red Line induced seismic signal at CSH, we are able to

[1] Note that if a significant signal were present at the location of Tres, we believe the statistical windowing would effectively capture the train induced vibration.

detect a strong increase in vibrational signal underneath the Red Line between 10 and 50 Hz. We hope our successes and failures in this endeavor facilitate future study of this noise at LUC, and that those studies are able to detect to much further distances.

VI. ACKNOWLEDGMENTS

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